

AHP MCDM Analysis Report

Native fuzzy AHP preserves triangular/trapezoidal pairwise judgments, derives fuzzy geometric means, defuzzifies priority weights, checks consistency on the defuzzified comparison matrix, and combines priorities with alternative performance.

Study Summary

Alternatives/factors: 5. Criteria/factors: 7. Top result: Alpha.

94 bundled numerical checks and 43 external published-example fixtures are included; TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Certification status: not publication-certified for every variant. Only registered passing external fixtures count as published-example validation evidence for the selected method and variant.

Study Mode Evidence

Input type: 35 fuzzy cells detected. Fuzzy types: triangular, trapezoidal. Mode: Native fuzzy AHP.

Respondent/group aggregation: No respondent matrix aggregation used. The uploaded/active decision matrix was treated as a single already aggregated dataset.

Expert aggregation: No DEMATEL expert matrix aggregation used. Only DEMATEL multiple-expert studies use expert direct-relation aggregation.

Fuzzy calculation convention: Native fuzzy AHP. Native fuzzy methods report the selected fuzzy mode; centroid-only methods report defuzzified input handling.

Validation Evidence

Certification status: Not publication-certified for every variant. Treat internal checks as implementation evidence. Treat only registered passing external fixtures as published-example validation for the selected method and variant.

Selected method validation status: Externally validated. 1 passing published-example fixture

Internal workflow and algorithm checks: Included. Registry integrity, method execution, group data, fuzzy input, zero-value handling, validation rules, templates, reports, and coverage summaries.

Bundled numerical benchmark checks: 94 checks. Known expected rankings and key coefficients for the built-in sample suites.

Selected method benchmark coverage: AHP. Selected method has bundled numerical benchmark coverage

External published-example validation: In progress: 43 fixtures. TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Selected method external validation: AHP. 1 external published-example fixture registered for this selected method: crisp-criteria-priority.

External fixture: AHP crisp-criteria-priority. Energies 2026 worked AHP example; DOI 10.3390/en19092214. AHP 3x3 criteria pairwise matrix, criteria priorities, lambda max, CI, CR, and consistency diagnostic. Source: <https://www.mdpi.com/1996-1073/19/9/2214>

Calculation Steps

1. Build criteria pairwise comparison matrix.
2. Normalize pairwise matrix and derive criteria priorities.
3. Calculate lambda max, CI, and CR.
4. Combine priorities with alternative performance scores.

Analyzed Decision Matrix

Alternative | C1 | C2 | C3 | C4 | C5 | C6 | C7

Alpha | 6 | 8.5 | 10 | 12.5 | 14 | 16.5 | 18

Beta | 9 | 12.5 | 15 | 18.5 | 21 | 24.5 | 27
Gamma | 12 | 16.5 | 20 | 24.5 | 28 | 32.5 | 36
Delta | 15 | 20.5 | 25 | 30.5 | 35 | 40.5 | 45
Epsilon | 18 | 24.5 | 30 | 36.5 | 42 | 48.5 | 54

Fuzzy AHP Aggregation Summary

Judgment type | Fuzzy matrices | Priority method
Criteria pairwise | 1 | Fuzzy geometric mean with centroid priority
Alternative pairwise | 7 | Fuzzy geometric mean by criterion

Fuzzy Criteria Pairwise Matrix

Criterion | C1 | C2 | C3 | C4 | C5 | C6 | C7
C1 | (1, 1, 1) | (1, 1, 1) | (2, 2, 2) | (1, 1, 1) | (2, 2, 2) | (2, 2, 2) | (3, 3, 3)
C2 | (1, 1, 1) | (1, 1, 1) | (2, 2, 2) | (1, 1, 1) | (2, 2, 2) | (2, 2, 2) | (3, 3, 3)
C3 | (0.5, 0.5, 0.5) | (0.5, 0.5, 0.5) | (1, 1, 1) | (0.5, 0.5, 0.5) | (1, 1, 1) | (1, 1, 1) | (2, 2, 2)
C4 | (1, 1, 1) | (1, 1, 1) | (2, 2, 2) | (1, 1, 1) | (2, 2, 2) | (2, 2, 2) | (3, 3, 3)
C5 | (0.5, 0.5, 0.5) | (0.5, 0.5, 0.5) | (1, 1, 1) | (0.5, 0.5, 0.5) | (1, 1, 1) | (1, 1, 1) | (2, 2, 2)
C6 | (0.5, 0.5, 0.5) | (0.5, 0.5, 0.5) | (1, 1, 1) | (0.5, 0.5, 0.5) | (1, 1, 1) | (1, 1, 1) | (2, 2, 2)
C7 | (0.3333, 0.3333, 0.3333) | (0.3333, 0.3333, 0.3333) | (0.5, 0.5, 0.5) | (0.3333, 0.3333, 0.3333) | (0.5, 0.5, 0.5) | (0.5, 0.5, 0.5) | (1, 1, 1)

Fuzzy Criteria Geometric Means

Criterion | Fuzzy geometric mean | Defuzzified priority
C1 | (1.5746, 1.5746, 1.5746) | 0.2059
C2 | (1.5746, 1.5746, 1.5746) | 0.2059
C3 | (0.8203, 0.8203, 0.8203) | 0.1073
C4 | (1.5746, 1.5746, 1.5746) | 0.2059
C5 | (0.8203, 0.8203, 0.8203) | 0.1073
C6 | (0.8203, 0.8203, 0.8203) | 0.1073
C7 | (0.464, 0.464, 0.464) | 0.0607

Fuzzy AHP Criteria Priorities

Criterion | Name | Priority
C1 | Cost | 0.2059
C2 | Environmental Management | 0.2059
C3 | CO2 Emissions | 0.1073
C4 | Labor Practices | 0.2059
C5 | Innovation | 0.1073
C6 | Community Impact | 0.1073
C7 | Waste | 0.0607

Fuzzy AHP Alternative Priorities By Criterion

Alternative | C1 | C2 | C3 | C4 | C5 | C6 | C7
Alpha | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2
Beta | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2
Gamma | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2
Delta | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2
Epsilon | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2

Priority-Compatible Normalized Matrix

Alternative | C1 | C2 | C3 | C4 | C5 | C6 | C7
Alpha | 1 | 0.3469 | 1 | 0.3425 | 0.3333 | 0.3402 | 1
Beta | 0.6667 | 0.5102 | 0.6667 | 0.5068 | 0.5 | 0.5052 | 0.6667
Gamma | 0.5 | 0.6735 | 0.5 | 0.6712 | 0.6667 | 0.6701 | 0.5
Delta | 0.4 | 0.8367 | 0.4 | 0.8356 | 0.8333 | 0.8351 | 0.4
Epsilon | 0.3333 | 1 | 0.3333 | 1 | 1 | 1 | 0.3333

Weighted Priority Matrix

Alternative | C1 | C2 | C3 | C4 | C5 | C6 | C7
Alpha | 0.0412 | 0.0412 | 0.0215 | 0.0412 | 0.0215 | 0.0215 | 0.0121

Beta | 0.0412 | 0.0412 | 0.0215 | 0.0412 | 0.0215 | 0.0215 | 0.0121
Gamma | 0.0412 | 0.0412 | 0.0215 | 0.0412 | 0.0215 | 0.0215 | 0.0121
Delta | 0.0412 | 0.0412 | 0.0215 | 0.0412 | 0.0215 | 0.0215 | 0.0121
Epsilon | 0.0412 | 0.0412 | 0.0215 | 0.0412 | 0.0215 | 0.0215 | 0.0121

Fuzzy AHP Defuzzified Consistency Summary

Metric | Value
Lambda max | 7.0152
CI | 0.0025
CR | 0.0019
Threshold | 0.1
Status | Accepted

Applied Criteria Weights

Criterion | Name | Direction | Applied Weight
C1 | Cost | cost | 0.2059
C2 | Environmental Management | benefit | 0.2059
C3 | CO2 Emissions | cost | 0.1073
C4 | Labor Practices | benefit | 0.2059
C5 | Innovation | benefit | 0.1073
C6 | Community Impact | benefit | 0.1073
C7 | Waste | cost | 0.0607

Final Ranking

Rank | Alternative | Score
1 | Alpha | 0.2
2 | Beta | 0.2
3 | Gamma | 0.2
4 | Delta | 0.2
5 | Epsilon | 0.2

Sensitivity Analysis

Criterion | Name | Base Weight | Low Scenario | High Scenario | Top Alternative | Stability Note
C1 | Cost | 0.2059 | 0.1853 | 0.2264 | Alpha | Review with +/-10% weight perturbation
C2 | Environmental Management | 0.2059 | 0.1853 | 0.2264 | Alpha | Review with +/-10% weight perturbation
C3 | CO2 Emissions | 0.1073 | 0.0965 | 0.118 | Alpha | Review with +/-10% weight perturbation
C4 | Labor Practices | 0.2059 | 0.1853 | 0.2264 | Alpha | Review with +/-10% weight perturbation
C5 | Innovation | 0.1073 | 0.0965 | 0.118 | Alpha | Review with +/-10% weight perturbation
C6 | Community Impact | 0.1073 | 0.0965 | 0.118 | Alpha | Review with +/-10% weight perturbation
C7 | Waste | 0.0607 | 0.0546 | 0.0667 | Alpha | Review with +/-10% weight perturbation

Figure data: Ranking Bar Chart

alternative | score | rank
Alpha | 0.2 | 1
Beta | 0.2 | 2
Gamma | 0.2 | 3
Delta | 0.2 | 4
Epsilon | 0.2 | 5

Figure data: Criteria Weights

criterion | weight
C1 | 0.2059
C2 | 0.2059
C3 | 0.1073
C4 | 0.2059
C5 | 0.1073
C6 | 0.1073
C7 | 0.0607

Figure data: Priority-Compatible Normalized Matrix Heatmap

alternative | criterion | value
Alpha | C1 | 1
Alpha | C2 | 0.3469
Alpha | C3 | 1
Alpha | C4 | 0.3425
Alpha | C5 | 0.3333
Alpha | C6 | 0.3402
Alpha | C7 | 1
Beta | C1 | 0.6667
Beta | C2 | 0.5102
Beta | C3 | 0.6667
Beta | C4 | 0.5068
Beta | C5 | 0.5
Beta | C6 | 0.5052
Beta | C7 | 0.6667
Gamma | C1 | 0.5
Gamma | C2 | 0.6735
Gamma | C3 | 0.5
Gamma | C4 | 0.6712
Gamma | C5 | 0.6667
Gamma | C6 | 0.6701
Gamma | C7 | 0.5
Delta | C1 | 0.4
Delta | C2 | 0.8367
Delta | C3 | 0.4
Delta | C4 | 0.8356
Delta | C5 | 0.8333
Delta | C6 | 0.8351
Delta | C7 | 0.4
Epsilon | C1 | 0.3333
Epsilon | C2 | 1
Epsilon | C3 | 0.3333
Epsilon | C4 | 1
Epsilon | C5 | 1
Epsilon | C6 | 1
Epsilon | C7 | 0.3333

Figure data: Weight Perturbation Band

criterion | name | baseWeight | lowScenario | highScenario | topAlternative
C1 | Cost | 0.2059 | 0.1853 | 0.2264 | Alpha
C2 | Environmental Management | 0.2059 | 0.1853 | 0.2264 | Alpha
C3 | CO2 Emissions | 0.1073 | 0.0965 | 0.118 | Alpha
C4 | Labor Practices | 0.2059 | 0.1853 | 0.2264 | Alpha
C5 | Innovation | 0.1073 | 0.0965 | 0.118 | Alpha
C6 | Community Impact | 0.1073 | 0.0965 | 0.118 | Alpha
C7 | Waste | 0.0607 | 0.0546 | 0.0667 | Alpha